and acting on these with the $\mathcal{R}\{\cdot\}$ operator, we obtain expressions for the elements of the vector $\mathbf{v}^{\mathrm{T}}\mathbf{H}$

$$\mathcal{R}\left\{\frac{\partial E}{\partial w_{kj}}\right\} = \mathcal{R}\{\delta_k\}z_j + \delta_k\mathcal{R}\{z_j\} \tag{5.110}$$

$$\mathcal{R}\left\{\frac{\partial E}{\partial w_{ji}}\right\} = x_i\mathcal{R}\{\delta_j\}. \tag{5.111}$$

The implementation of this algorithm involves the introduction of additional variables $\mathcal{R}\{a_j\}$, $\mathcal{R}\{z_j\}$ and $\mathcal{R}\{\delta_j\}$ for the hidden units and $\mathcal{R}\{\delta_k\}$ and $\mathcal{R}\{y_k\}$ for the output units. For each input pattern, the values of these quantities can be found using the above results, and the elements of $\mathbf{v}^{\mathrm{T}}\mathbf{H}$ are then given by (5.110) and (5.111). An elegant aspect of this technique is that the equations for evaluating $\mathbf{v}^{\mathrm{T}}\mathbf{H}$ mirror closely those for standard forward and backward propagation, and so the extension of existing software to compute this product is typically straightforward.

If desired, the technique can be used to evaluate the full Hessian matrix by choosing the vector $\mathbf{v}$ to be given successively by a series of unit vectors of the form $(0, 0, \ldots, 1, \ldots, 0)$ each of which picks out one column of the Hessian. This leads to a formalism that is analytically equivalent to the backpropagation procedure of Bishop (1992), as described in Section 5.4.5, though with some loss of efficiency due to redundant calculations.

5.5. Regularization in Neural Networks

The number of input and outputs units in a neural network is generally determined by the dimensionality of the data set, whereas the number M of hidden units is a free parameter that can be adjusted to give the best predictive performance. Note that M controls the number of parameters (weights and biases) in the network, and so we might expect that in a maximum likelihood setting there will be an optimum value of M that gives the best generalization performance, corresponding to the optimum balance between under-fitting and over-fitting. Figure 5.9 shows an example of the effect of different values of M for the sinusoidal regression problem.

The generalization error, however, is not a simple function of M due to the presence of local minima in the error function, as illustrated in Figure 5.10. Here we see the effect of choosing multiple random initializations for the weight vector for a range of values of M. The overall best validation set performance in this case occurred for a particular solution having $M = 8$. In practice, one approach to choosing M is in fact to plot a graph of the kind shown in Figure 5.10 and then to choose the specific solution having the smallest validation set error.

There are, however, other ways to control the complexity of a neural network model in order to avoid over-fitting. From our discussion of polynomial curve fitting in Chapter 1, we see that an alternative approach is to choose a relatively large value for M and then to control complexity by the addition of a regularization term to the error function. The simplest regularizer is the quadratic, giving a regularized error

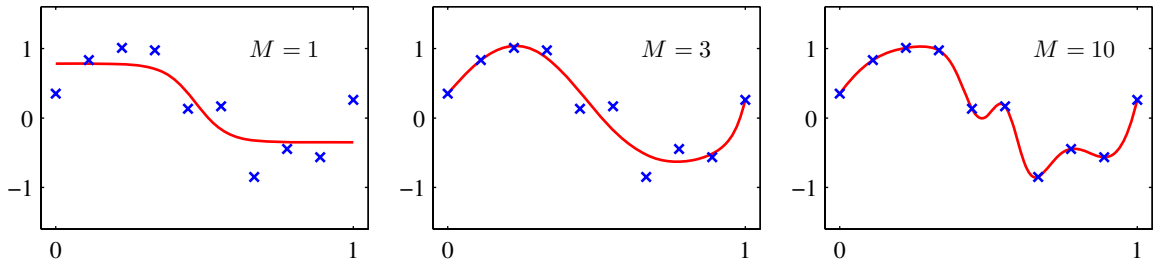


Figure 5.9 Examples of two-layer networks trained on 10 data points drawn from the sinusoidal data set. The graphs show the result of fitting networks having $M = 1$, 3 and 10 hidden units, respectively, by minimizing a sum-of-squares error function using a scaled conjugate-gradient algorithm.

of the form

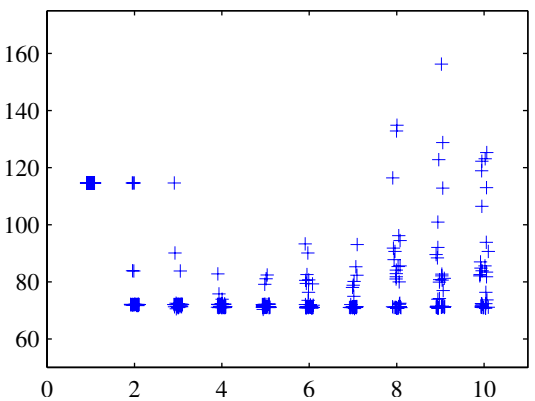
$$\widetilde{E}(\mathbf{w}) = E(\mathbf{w}) + \frac{\lambda}{2}\mathbf{w}^{\mathrm{T}}\mathbf{w}. \tag{5.112}$$

This regularizer is also known as *weight decay* and has been discussed at length in Chapter 3. The effective model complexity is then determined by the choice of the regularization coefficient λ. As we have seen previously, this regularizer can be interpreted as the negative logarithm of a zero-mean Gaussian prior distribution over the weight vector $\mathbf{w}$.

5.5.1 Consistent Gaussian priors

One of the limitations of simple weight decay in the form (5.112) is that is inconsistent with certain scaling properties of network mappings. To illustrate this, consider a multilayer perceptron network having two layers of weights and linear output units, which performs a mapping from a set of input variables $\{x_i\}$ to a set of output variables $\{y_k\}$. The activations of the hidden units in the first hidden layer

Figure 5.10 Plot of the sum-of-squares test-set error for the polynomial data set versus the number of hidden units in the network, with 30 random starts for each network size, showing the effect of local minima. For each new start, the weight vector was initialized by sampling from an isotropic Gaussian distribution having a mean of zero and a variance of 10.



take the form

$$z_j = h\left(\sum_i w_{ji}x_i + w_{j0}\right) \tag{5.113}$$

while the activations of the output units are given by

$$y_k = \sum_j w_{kj}z_j + w_{k0}. \tag{5.114}$$

Suppose we perform a linear transformation of the input data of the form

$$x_i \to \widetilde{x}_i = ax_i + b. \tag{5.115}$$

Then we can arrange for the mapping performed by the network to be unchanged by making a corresponding linear transformation of the weights and biases from the inputs to the units in the hidden layer of the form

Exercise 5.24

$$w_{ji} \to \widetilde{w}_{ji} = \frac{1}{a}w_{ji} \tag{5.116}$$

$$w_{j0} \to \widetilde{w}_{j0} = w_{j0} - \frac{b}{a}\sum_i w_{ji}. \tag{5.117}$$

Similarly, a linear transformation of the output variables of the network of the form

$$y_k \to \widetilde{y}_k = cy_k + d \tag{5.118}$$

can be achieved by making a transformation of the second-layer weights and biases using

$$w_{kj} \to \widetilde{w}_{kj} = cw_{kj} \tag{5.119}$$

$$w_{k0} \to \widetilde{w}_{k0} = cw_{k0} + d. \tag{5.120}$$

If we train one network using the original data and one network using data for which the input and/or target variables are transformed by one of the above linear transformations, then consistency requires that we should obtain equivalent networks that differ only by the linear transformation of the weights as given. Any regularizer should be consistent with this property, otherwise it arbitrarily favours one solution over another, equivalent one. Clearly, simple weight decay (5.112), that treats all weights and biases on an equal footing, does not satisfy this property.

We therefore look for a regularizer which is invariant under the linear transformations (5.116), (5.117), (5.119) and (5.120). These require that the regularizer should be invariant to re-scaling of the weights and to shifts of the biases. Such a regularizer is given by

$$\frac{\lambda_1}{2}\sum_{w\in\mathcal{W}_1} w^2 + \frac{\lambda_2}{2}\sum_{w\in\mathcal{W}_2} w^2 \tag{5.121}$$

where $\mathcal{W}_1$ denotes the set of weights in the first layer, $\mathcal{W}_2$ denotes the set of weights in the second layer, and biases are excluded from the summations. This regularizer

will remain unchanged under the weight transformations provided the regularization parameters are re-scaled using $\lambda_1 \to a^{1/2}\lambda_1$ and $\lambda_2 \to c^{-1/2}\lambda_2$.

The regularizer (5.121) corresponds to a prior of the form

$$p(\mathbf{w}|\alpha_1, \alpha_2) \propto \exp\left(-\frac{\alpha_1}{2}\sum_{w\in\mathcal{W}_1} w^2 - \frac{\alpha_2}{2}\sum_{w\in\mathcal{W}_2} w^2\right). \tag{5.122}$$

Note that priors of this form are *improper* (they cannot be normalized) because the bias parameters are unconstrained. The use of improper priors can lead to difficulties in selecting regularization coefficients and in model comparison within the Bayesian framework, because the corresponding evidence is zero. It is therefore common to include separate priors for the biases (which then break shift invariance) having their own hyperparameters. We can illustrate the effect of the resulting four hyperparameters by drawing samples from the prior and plotting the corresponding network functions, as shown in Figure 5.11.

More generally, we can consider priors in which the weights are divided into any number of groups $\mathcal{W}_k$ so that

$$p(\mathbf{w}) \propto \exp\left(-\frac{1}{2}\sum_k \alpha_k \|\mathbf{w}\|_k^2\right) \tag{5.123}$$

where

$$\|\mathbf{w}\|_k^2 = \sum_{j\in\mathcal{W}_k} w_j^2. \tag{5.124}$$

As a special case of this prior, if we choose the groups to correspond to the sets of weights associated with each of the input units, and we optimize the marginal likelihood with respect to the corresponding parameters α_k, we obtain *automatic relevance determination* as discussed in Section 7.2.2.

5.5.2 Early stopping

An alternative to regularization as a way of controlling the effective complexity of a network is the procedure of *early stopping*. The training of nonlinear network models corresponds to an iterative reduction of the error function defined with respect to a set of training data. For many of the optimization algorithms used for network training, such as conjugate gradients, the error is a nonincreasing function of the iteration index. However, the error measured with respect to independent data, generally called a validation set, often shows a decrease at first, followed by an increase as the network starts to over-fit. Training can therefore be stopped at the point of smallest error with respect to the validation data set, as indicated in Figure 5.12, in order to obtain a network having good generalization performance.

The behaviour of the network in this case is sometimes explained qualitatively in terms of the effective number of degrees of freedom in the network, in which this number starts out small and then to grows during the training process, corresponding to a steady increase in the effective complexity of the model. Halting training before

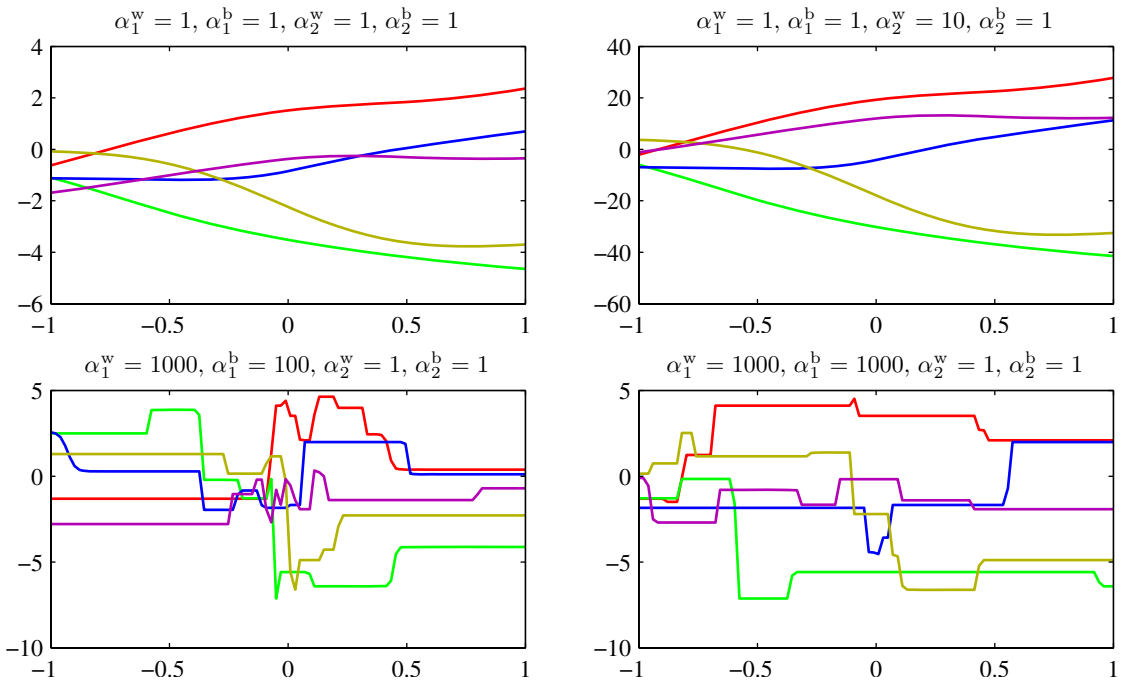


Figure 5.11 Illustration of the effect of the hyperparameters governing the prior distribution over weights and biases in a two-layer network having a single input, a single linear output, and 12 hidden units having 'tanh' activation functions. The priors are governed by four hyperparameters α_1^{b}, α_1^{w}, α_2^{b}, and α_2^{w}, which represent the precisions of the Gaussian distributions of the first-layer biases, first-layer weights, second-layer biases, and second-layer weights, respectively. We see that the parameter α_2^{w} governs the vertical scale of functions (note the different vertical axis ranges on the top two diagrams), α_1^{w} governs the horizontal scale of variations in the function values, and α_1^{b} governs the horizontal range over which variations occur. The parameter α_2^{b}, whose effect is not illustrated here, governs the range of vertical offsets of the functions.

a minimum of the training error has been reached then represents a way of limiting the effective network complexity.

In the case of a quadratic error function, we can verify this insight, and show that early stopping should exhibit similar behaviour to regularization using a simple weight-decay term. This can be understood from Figure 5.13, in which the axes in weight space have been rotated to be parallel to the eigenvectors of the Hessian matrix. If, in the absence of weight decay, the weight vector starts at the origin and proceeds during training along a path that follows the local negative gradient vector, then the weight vector will move initially parallel to the w_2 axis through a point corresponding roughly to $\widetilde{\mathbf{w}}$ and then move towards the minimum of the error function $\mathbf{w}_{\mathrm{ML}}$. This follows from the shape of the error surface and the widely differing eigenvalues of the Hessian. Stopping at a point near $\widetilde{\mathbf{w}}$ is therefore similar to weight decay. The relationship between early stopping and weight decay can be made quantitative, thereby showing that the quantity $\tau\eta$ (where τ is the iteration index, and η is the learning rate parameter) plays the role of the reciprocal of the regularization

Exercise 5.25

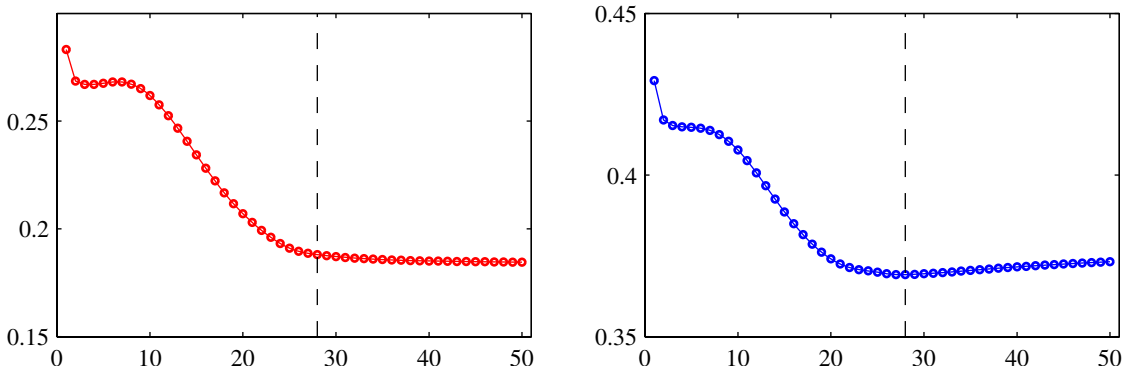


Figure 5.12 An illustration of the behaviour of training set error (left) and validation set error (right) during a typical training session, as a function of the iteration step, for the sinusoidal data set. The goal of achieving the best generalization performance suggests that training should be stopped at the point shown by the vertical dashed lines, corresponding to the minimum of the validation set error.

parameter λ. The effective number of parameters in the network therefore grows during the course of training.

5.5.3 Invariances

In many applications of pattern recognition, it is known that predictions should be unchanged, or *invariant*, under one or more transformations of the input variables. For example, in the classification of objects in two-dimensional images, such as handwritten digits, a particular object should be assigned the same classification irrespective of its position within the image (*translation invariance*) or of its size (*scale invariance*). Such transformations produce significant changes in the raw data, expressed in terms of the intensities at each of the pixels in the image, and yet should give rise to the same output from the classification system. Similarly in speech recognition, small levels of nonlinear warping along the time axis, which preserve temporal ordering, should not change the interpretation of the signal.

If sufficiently large numbers of training patterns are available, then an adaptive model such as a neural network can learn the invariance, at least approximately. This involves including within the training set a sufficiently large number of examples of the effects of the various transformations. Thus, for translation invariance in an image, the training set should include examples of objects at many different positions.

This approach may be impractical, however, if the number of training examples is limited, or if there are several invariants (because the number of combinations of transformations grows exponentially with the number of such transformations). We therefore seek alternative approaches for encouraging an adaptive model to exhibit the required invariances. These can broadly be divided into four categories:

1. The training set is augmented using replicas of the training patterns, transformed according to the desired invariances. For instance, in our digit recognition example, we could make multiple copies of each example in which the

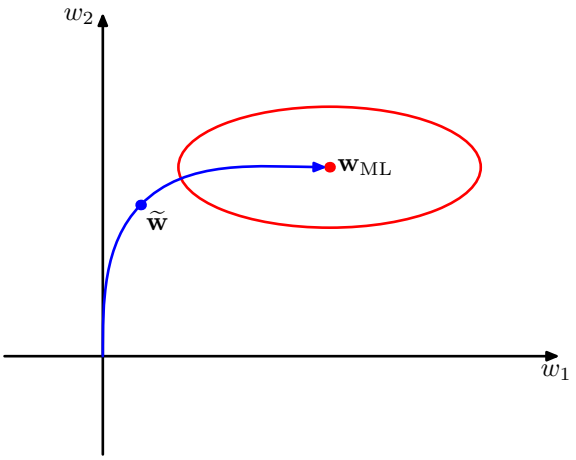


Figure 5.13 A schematic illustration of why early stopping can give similar results to weight decay in the case of a quadratic error function. The ellipse shows a contour of constant error, and $\mathbf{w}_{\mathrm{ML}}$ denotes the minimum of the error function. If the weight vector starts at the origin and moves according to the local negative gradient direction, then it will follow the path shown by the curve. By stopping training early, a weight vector $\widetilde{\mathbf{w}}$ is found that is qualitatively similar to that obtained with a simple weight-decay regularizer and training to the minimum of the regularized error, as can be seen by comparing with Figure 3.15.

digit is shifted to a different position in each image.

2. A regularization term is added to the error function that penalizes changes in the model output when the input is transformed. This leads to the technique of *tangent propagation*, discussed in Section 5.5.4.

3. Invariance is built into the pre-processing by extracting features that are invariant under the required transformations. Any subsequent regression or classification system that uses such features as inputs will necessarily also respect these invariances.

4. The final option is to build the invariance properties into the structure of a neural network (or into the definition of a kernel function in the case of techniques such as the relevance vector machine). One way to achieve this is through the use of local receptive fields and shared weights, as discussed in the context of convolutional neural networks in Section 5.5.6.

Approach 1 is often relatively easy to implement and can be used to encourage complex invariances such as those illustrated in Figure 5.14. For sequential training algorithms, this can be done by transforming each input pattern before it is presented to the model so that, if the patterns are being recycled, a different transformation (drawn from an appropriate distribution) is added each time. For batch methods, a similar effect can be achieved by replicating each data point a number of times and transforming each copy independently. The use of such augmented data can lead to significant improvements in generalization (Simard *et al.*, 2003), although it can also be computationally costly.

Approach 2 leaves the data set unchanged but modifies the error function through the addition of a regularizer. In Section 5.5.5, we shall show that this approach is closely related to approach 2.

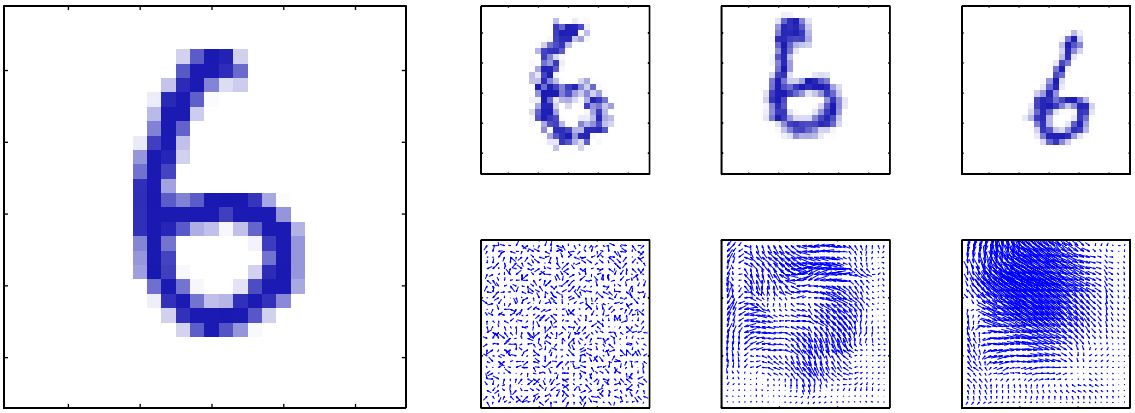


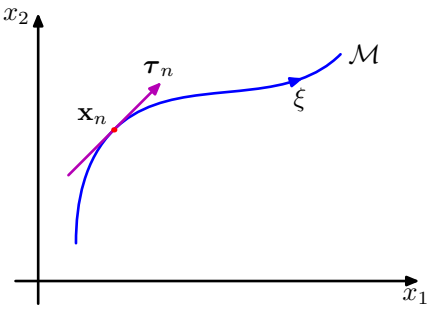
Figure 5.14 Illustration of the synthetic warping of a handwritten digit. The original image is shown on the left. On the right, the top row shows three examples of warped digits, with the corresponding displacement fields shown on the bottom row. These displacement fields are generated by sampling random displacements $\Delta x, \Delta y \in (0, 1)$ at each pixel and then smoothing by convolution with Gaussians of width 0.01, 30 and 60 respectively.

One advantage of approach 3 is that it can correctly extrapolate well beyond the range of transformations included in the training set. However, it can be difficult to find hand-crafted features with the required invariances that do not also discard information that can be useful for discrimination.

5.5.4 Tangent propagation

We can use regularization to encourage models to be invariant to transformations of the input through the technique of *tangent propagation* (Simard *et al.*, 1992). Consider the effect of a transformation on a particular input vector $\mathbf{x}_n$. Provided the transformation is continuous (such as translation or rotation, but not mirror reflection for instance), then the transformed pattern will sweep out a manifold $\mathcal{M}$ within the D-dimensional input space. This is illustrated in Figure 5.15, for the case of $D = 2$ for simplicity. Suppose the transformation is governed by a single parameter ξ (which might be rotation angle for instance). Then the subspace $\mathcal{M}$ swept out by $\mathbf{x}_n$

Figure 5.15 Illustration of a two-dimensional input space showing the effect of a continuous transformation on a particular input vector $\mathbf{x}_n$. A one-dimensional transformation, parameterized by the continuous variable ξ, applied to $\mathbf{x}_n$ causes it to sweep out a one-dimensional manifold $\mathcal{M}$. Locally, the effect of the transformation can be approximated by the tangent vector $\boldsymbol{\tau}_n$.



will be one-dimensional, and will be parameterized by ξ. Let the vector that results from acting on $\mathbf{x}_n$ by this transformation be denoted by $\mathbf{s}(\mathbf{x}_n, \xi)$, which is defined so that $\mathbf{s}(\mathbf{x}, 0) = \mathbf{x}$. Then the tangent to the curve $\mathcal{M}$ is given by the directional derivative $\boldsymbol{\tau} = \partial\mathbf{s}/\partial\xi$, and the tangent vector at the point $\mathbf{x}_n$ is given by

$$\boldsymbol{\tau}_n = \left.\frac{\partial \mathbf{s}(\mathbf{x}_n, \xi)}{\partial \xi}\right|_{\xi=0}. \tag{5.125}$$

Under a transformation of the input vector, the network output vector will, in general, change. The derivative of output k with respect to ξ is given by

$$\left.\frac{\partial y_k}{\partial \xi}\right|_{\xi=0} = \sum_{i=1}^{D} \frac{\partial y_k}{\partial x_i}\frac{\partial x_i}{\partial \xi}\bigg|_{\xi=0} = \sum_{i=1}^{D} J_{ki}\tau_i \tag{5.126}$$

where J_{ki} is the (k, i) element of the Jacobian matrix $\mathbf{J}$, as discussed in Section 5.3.4. The result (5.126) can be used to modify the standard error function, so as to encourage local invariance in the neighbourhood of the data points, by the addition to the original error function E of a regularization function Ω to give a total error function of the form

$$\widetilde{E} = E + \lambda\Omega \tag{5.127}$$

where λ is a regularization coefficient and

$$\Omega = \frac{1}{2}\sum_n \sum_k \left(\left.\frac{\partial y_{nk}}{\partial \xi}\right|_{\xi=0}\right)^2 = \frac{1}{2}\sum_n \sum_k \left(\sum_{i=1}^{D} J_{nki}\tau_{ni}\right)^2. \tag{5.128}$$

The regularization function will be zero when the network mapping function is invariant under the transformation in the neighbourhood of each pattern vector, and the value of the parameter λ determines the balance between fitting the training data and learning the invariance property.

In a practical implementation, the tangent vector $\boldsymbol{\tau}_n$ can be approximated using finite differences, by subtracting the original vector $\mathbf{x}_n$ from the corresponding vector after transformation using a small value of ξ, and then dividing by ξ. This is illustrated in Figure 5.16.

The regularization function depends on the network weights through the Jacobian $\mathbf{J}$. A backpropagation formalism for computing the derivatives of the regularizer with respect to the network weights is easily obtained by extension of the techniques introduced in Section 5.3.

Exercise 5.26

If the transformation is governed by L parameters (e.g., $L = 3$ for the case of translations combined with in-plane rotations in a two-dimensional image), then the manifold $\mathcal{M}$ will have dimensionality L, and the corresponding regularizer is given by the sum of terms of the form (5.128), one for each transformation. If several transformations are considered at the same time, and the network mapping is made invariant to each separately, then it will be (locally) invariant to combinations of the transformations (Simard *et al.*, 1992).

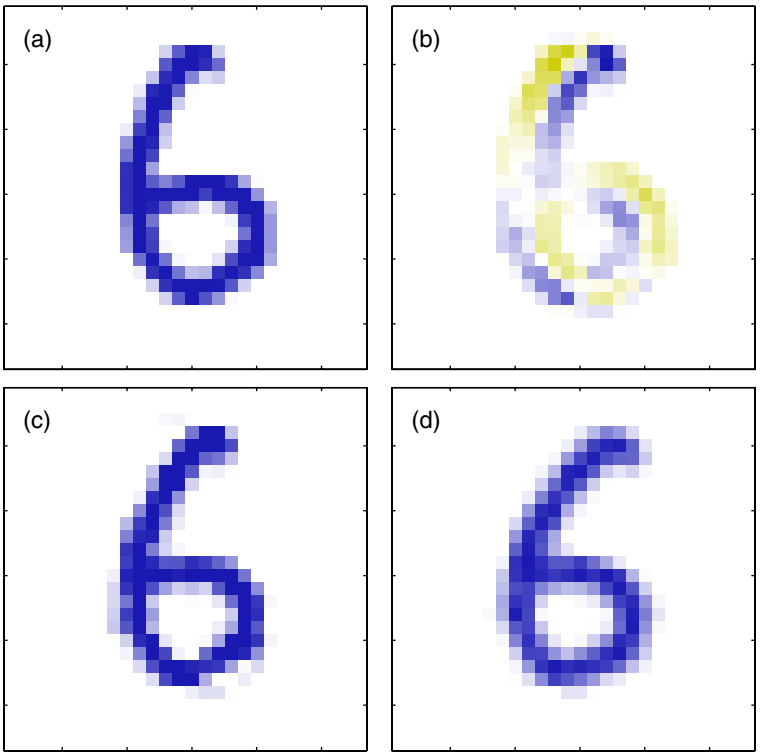


Figure 5.16 Illustration showing (a) the original image $\mathbf{x}$ of a handwritten digit, (b) the tangent vector $\boldsymbol{\tau}$ corresponding to an infinitesimal clockwise rotation, (c) the result of adding a small contribution from the tangent vector to the original image giving $\mathbf{x} + \epsilon\boldsymbol{\tau}$ with $\epsilon = 15$ degrees, and (d) the true image rotated for comparison.

A related technique, called *tangent distance*, can be used to build invariance properties into distance-based methods such as nearest-neighbour classifiers (Simard *et al.*, 1993).

5.5.5 Training with transformed data

We have seen that one way to encourage invariance of a model to a set of transformations is to expand the training set using transformed versions of the original input patterns. Here we show that this approach is closely related to the technique of tangent propagation (Bishop, 1995b; Leen, 1995).

As in Section 5.5.4, we shall consider a transformation governed by a single parameter ξ and described by the function $\mathbf{s}(\mathbf{x}, \xi)$, with $\mathbf{s}(\mathbf{x}, 0) = \mathbf{x}$. We shall also consider a sum-of-squares error function. The error function for untransformed inputs can be written (in the infinite data set limit) in the form

$$E = \frac{1}{2} \iint \{y(\mathbf{x}) - t\}^2 p(t|\mathbf{x}) p(\mathbf{x}) \, \mathrm{d}\mathbf{x} \, \mathrm{d}t \tag{5.129}$$

as discussed in Section 1.5.5. Here we have considered a network having a single output, in order to keep the notation uncluttered. If we now consider an infinite number of copies of each data point, each of which is perturbed by the transformation

in which the parameter ξ is drawn from a distribution $p(\xi)$, then the error function defined over this expanded data set can be written as

$$\widetilde{E} = \frac{1}{2}\iiint \{y(\mathbf{s}(\mathbf{x},\xi)) - t\}^2 p(t|\mathbf{x})p(\mathbf{x})p(\xi)\,\mathrm{d}\mathbf{x}\,\mathrm{d}t\,\mathrm{d}\xi. \tag{5.130}$$

We now assume that the distribution $p(\xi)$ has zero mean with small variance, so that we are only considering small transformations of the original input vectors. We can then expand the transformation function as a Taylor series in powers of ξ to give

$$\begin{aligned}\mathbf{s}(\mathbf{x},\xi) &= \mathbf{s}(\mathbf{x},0) + \xi\,\frac{\partial}{\partial\xi}\mathbf{s}(\mathbf{x},\xi)\bigg|_{\xi=0} + \frac{\xi^2}{2}\,\frac{\partial^2}{\partial\xi^2}\mathbf{s}(\mathbf{x},\xi)\bigg|_{\xi=0} + O(\xi^3)\\ &= \mathbf{x} + \xi\boldsymbol{\tau} + \frac{1}{2}\xi^2\boldsymbol{\tau}' + O(\xi^3)\end{aligned}$$

where $\boldsymbol{\tau}'$ denotes the second derivative of $\mathbf{s}(\mathbf{x},\xi)$ with respect to ξ evaluated at $\xi = 0$. This allows us to expand the model function to give

$$y(\mathbf{s}(\mathbf{x},\xi)) = y(\mathbf{x}) + \xi\boldsymbol{\tau}^{\mathrm{T}}\nabla y(\mathbf{x}) + \frac{\xi^2}{2}\left[(\boldsymbol{\tau}')^{\mathrm{T}}\,\nabla y(\mathbf{x}) + \boldsymbol{\tau}^{\mathrm{T}}\nabla\nabla y(\mathbf{x})\boldsymbol{\tau}\right] + O(\xi^3).$$

Substituting into the mean error function (5.130) and expanding, we then have

$$\begin{aligned}\widetilde{E} &= \frac{1}{2}\iint \{y(\mathbf{x}) - t\}^2 p(t|\mathbf{x})p(\mathbf{x})\,\mathrm{d}\mathbf{x}\,\mathrm{d}t\\ &+ \mathbb{E}[\xi]\iint \{y(\mathbf{x}) - t\}\boldsymbol{\tau}^{\mathrm{T}}\nabla y(\mathbf{x})p(t|\mathbf{x})p(\mathbf{x})\,\mathrm{d}\mathbf{x}\,\mathrm{d}t\\ &+ \mathbb{E}[\xi^2]\iint \left[\{y(\mathbf{x}) - t\}\frac{1}{2}\left\{(\boldsymbol{\tau}')^{\mathrm{T}}\,\nabla y(\mathbf{x}) + \boldsymbol{\tau}^{\mathrm{T}}\nabla\nabla y(\mathbf{x})\boldsymbol{\tau}\right\}\right.\\ &\qquad \left. + \left(\boldsymbol{\tau}^{\mathrm{T}}\nabla y(\mathbf{x})\right)^2\right] p(t|\mathbf{x})p(\mathbf{x})\,\mathrm{d}\mathbf{x}\,\mathrm{d}t + O(\xi^3).\end{aligned}$$

Because the distribution of transformations has zero mean we have $\mathbb{E}[\xi] = 0$. Also, we shall denote $\mathbb{E}[\xi^2]$ by λ. Omitting terms of $O(\xi^3)$, the average error function then becomes

$$\widetilde{E} = E + \lambda\Omega \tag{5.131}$$

where E is the original sum-of-squares error, and the regularization term Ω takes the form

$$\begin{aligned}\Omega &= \int \left[\{y(\mathbf{x}) - \mathbb{E}[t|\mathbf{x}]\}\frac{1}{2}\left\{(\boldsymbol{\tau}')^{\mathrm{T}}\,\nabla y(\mathbf{x}) + \boldsymbol{\tau}^{\mathrm{T}}\nabla\nabla y(\mathbf{x})\boldsymbol{\tau}\right\}\right.\\ &\qquad \left. + \left(\boldsymbol{\tau}^{T}\nabla y(\mathbf{x})\right)^2\right] p(\mathbf{x})\,\mathrm{d}\mathbf{x}\end{aligned} \tag{5.132}$$

in which we have performed the integration over t.

We can further simplify this regularization term as follows. In Section 1.5.5 we saw that the function that minimizes the sum-of-squares error is given by the conditional average $\mathbb{E}[t|\mathbf{x}]$ of the target values t. From (5.131) we see that the regularized error will equal the unregularized sum-of-squares plus terms which are $O(\xi)$, and so the network function that minimizes the total error will have the form

$$y(\mathbf{x}) = \mathbb{E}[t|\mathbf{x}] + O(\xi). \tag{5.133}$$

Thus, to leading order in ξ, the first term in the regularizer vanishes and we are left with

$$\Omega = \frac{1}{2}\int \left(\boldsymbol{\tau}^T \nabla y(\mathbf{x})\right)^2 p(\mathbf{x})\,\mathrm{d}\mathbf{x} \tag{5.134}$$

which is equivalent to the tangent propagation regularizer (5.128).

If we consider the special case in which the transformation of the inputs simply consists of the addition of random noise, so that $\mathbf{x} \to \mathbf{x} + \boldsymbol{\xi}$, then the regularizer takes the form

Exercise 5.27

$$\Omega = \frac{1}{2}\int \|\nabla y(\mathbf{x})\|^2 p(\mathbf{x})\,\mathrm{d}\mathbf{x} \tag{5.135}$$

which is known as *Tikhonov* regularization (Tikhonov and Arsenin, 1977; Bishop, 1995b). Derivatives of this regularizer with respect to the network weights can be found using an extended backpropagation algorithm (Bishop, 1993). We see that, for small noise amplitudes, Tikhonov regularization is related to the addition of random noise to the inputs, which has been shown to improve generalization in appropriate circumstances (Sietsma and Dow, 1991).

5.5.6 Convolutional networks

Another approach to creating models that are invariant to certain transformation of the inputs is to build the invariance properties into the structure of a neural network. This is the basis for the *convolutional neural network* (Le Cun *et al.*, 1989; LeCun *et al.*, 1998), which has been widely applied to image data.

Consider the specific task of recognizing handwritten digits. Each input image comprises a set of pixel intensity values, and the desired output is a posterior probability distribution over the ten digit classes. We know that the identity of the digit is invariant under translations and scaling as well as (small) rotations. Furthermore, the network must also exhibit invariance to more subtle transformations such as elastic deformations of the kind illustrated in Figure 5.14. One simple approach would be to treat the image as the input to a fully connected network, such as the kind shown in Figure 5.1. Given a sufficiently large training set, such a network could in principle yield a good solution to this problem and would learn the appropriate invariances by example.

However, this approach ignores a key property of images, which is that nearby pixels are more strongly correlated than more distant pixels. Many of the modern approaches to computer vision exploit this property by extracting *local* features that depend only on small subregions of the image. Information from such features can then be merged in later stages of processing in order to detect higher-order features

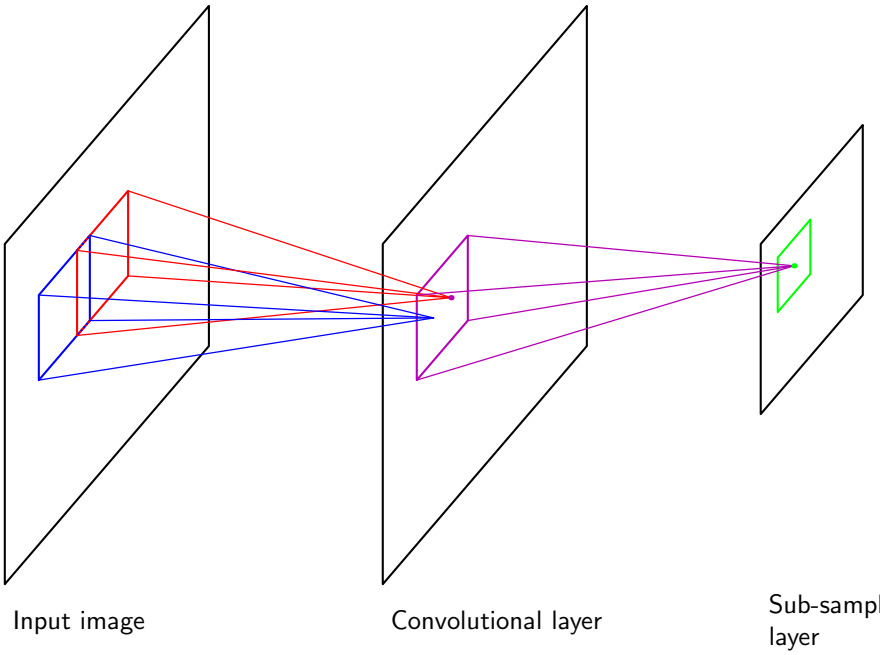


Figure 5.17 Diagram illustrating part of a convolutional neural network, showing a layer of convolutional units followed by a layer of subsampling units. Several successive pairs of such layers may be used.

and ultimately to yield information about the image as whole. Also, local features that are useful in one region of the image are likely to be useful in other regions of the image, for instance if the object of interest is translated.

These notions are incorporated into convolutional neural networks through three mechanisms: (i) local receptive fields, (ii) weight sharing, and (iii) subsampling. The structure of a convolutional network is illustrated in Figure 5.17. In the convolutional layer the units are organized into planes, each of which is called a *feature map*. Units in a feature map each take inputs only from a small subregion of the image, and all of the units in a feature map are constrained to share the same weight values. For instance, a feature map might consist of 100 units arranged in a 10×10 grid, with each unit taking inputs from a 5×5 pixel patch of the image. The whole feature map therefore has 25 adjustable weight parameters plus one adjustable bias parameter. Input values from a patch are linearly combined using the weights and the bias, and the result transformed by a sigmoidal nonlinearity using (5.1). If we think of the units as feature detectors, then all of the units in a feature map detect the same pattern but at different locations in the input image. Due to the weight sharing, the evaluation of the activations of these units is equivalent to a convolution of the image pixel intensities with a 'kernel' comprising the weight parameters. If the input image is shifted, the activations of the feature map will be shifted by the same amount but will otherwise be unchanged. This provides the basis for the (approximate) invariance of

the network outputs to translations and distortions of the input image. Because we will typically need to detect multiple features in order to build an effective model, there will generally be multiple feature maps in the convolutional layer, each having its own set of weight and bias parameters.

The outputs of the convolutional units form the inputs to the subsampling layer of the network. For each feature map in the convolutional layer, there is a plane of units in the subsampling layer and each unit takes inputs from a small receptive field in the corresponding feature map of the convolutional layer. These units perform subsampling. For instance, each subsampling unit might take inputs from a 2×2 unit region in the corresponding feature map and would compute the average of those inputs, multiplied by an adaptive weight with the addition of an adaptive bias parameter, and then transformed using a sigmoidal nonlinear activation function. The receptive fields are chosen to be contiguous and nonoverlapping so that there are half the number of rows and columns in the subsampling layer compared with the convolutional layer. In this way, the response of a unit in the subsampling layer will be relatively insensitive to small shifts of the image in the corresponding regions of the input space.

In a practical architecture, there may be several pairs of convolutional and subsampling layers. At each stage there is a larger degree of invariance to input transformations compared to the previous layer. There may be several feature maps in a given convolutional layer for each plane of units in the previous subsampling layer, so that the gradual reduction in spatial resolution is then compensated by an increasing number of features. The final layer of the network would typically be a fully connected, fully adaptive layer, with a softmax output nonlinearity in the case of multiclass classification.

The whole network can be trained by error minimization using backpropagation to evaluate the gradient of the error function. This involves a slight modification of the usual backpropagation algorithm to ensure that the shared-weight constraints are satisfied. Due to the use of local receptive fields, the number of weights in the network is smaller than if the network were fully connected. Furthermore, the number of independent parameters to be learned from the data is much smaller still, due to the substantial numbers of constraints on the weights.

Exercise 5.28

5.5.7 Soft weight sharing

One way to reduce the effective complexity of a network with a large number of weights is to constrain weights within certain groups to be equal. This is the technique of weight sharing that was discussed in Section 5.5.6 as a way of building translation invariance into networks used for image interpretation. It is only applicable, however, to particular problems in which the form of the constraints can be specified in advance. Here we consider a form of *soft weight sharing* (Nowlan and Hinton, 1992) in which the hard constraint of equal weights is replaced by a form of regularization in which groups of weights are encouraged to have similar values. Furthermore, the division of weights into groups, the mean weight value for each group, and the spread of values within the groups are all determined as part of the learning process.

Recall that the simple weight decay regularizer, given in (5.112), can be viewed as the negative log of a Gaussian prior distribution over the weights. We can encourage the weight values to form several groups, rather than just one group, by considering instead a probability distribution that is a *mixture* of Gaussians. The centres and variances of the Gaussian components, as well as the mixing coefficients, will be considered as adjustable parameters to be determined as part of the learning process. Thus, we have a probability density of the form

Section 2.3.9

$$p(\mathbf{w}) = \prod_i p(w_i) \tag{5.136}$$

where

$$p(w_i) = \sum_{j=1}^{M} \pi_j \mathcal{N}(w_i|\mu_j, \sigma_j^2) \tag{5.137}$$

and π_j are the mixing coefficients. Taking the negative logarithm then leads to a regularization function of the form

$$\Omega(\mathbf{w}) = -\sum_i \ln \left(\sum_{j=1}^{M} \pi_j \mathcal{N}(w_i|\mu_j, \sigma_j^2) \right). \tag{5.138}$$

The total error function is then given by

$$\widetilde{E}(\mathbf{w}) = E(\mathbf{w}) + \lambda \Omega(\mathbf{w}) \tag{5.139}$$

where λ is the regularization coefficient. This error is minimized both with respect to the weights w_i and with respect to the parameters $\{\pi_j, \mu_j, \sigma_j\}$ of the mixture model. If the weights were constant, then the parameters of the mixture model could be determined by using the EM algorithm discussed in Chapter 9. However, the distribution of weights is itself evolving during the learning process, and so to avoid numerical instability, a joint optimization is performed simultaneously over the weights and the mixture-model parameters. This can be done using a standard optimization algorithm such as conjugate gradients or quasi-Newton methods.

In order to minimize the total error function, it is necessary to be able to evaluate its derivatives with respect to the various adjustable parameters. To do this it is convenient to regard the $\{\pi_j\}$ as *prior* probabilities and to introduce the corresponding posterior probabilities which, following (2.192), are given by Bayes' theorem in the form

$$\gamma_j(w) = \frac{\pi_j \mathcal{N}(w|\mu_j, \sigma_j^2)}{\sum_k \pi_k \mathcal{N}(w|\mu_k, \sigma_k^2)}. \tag{5.140}$$

The derivatives of the total error function with respect to the weights are then given by

Exercise 5.29

$$\frac{\partial \widetilde{E}}{\partial w_i} = \frac{\partial E}{\partial w_i} + \lambda \sum_j \gamma_j(w_i) \frac{(w_i - \mu_j)}{\sigma_j^2}. \tag{5.141}$$

The effect of the regularization term is therefore to pull each weight towards the centre of the jth Gaussian, with a force proportional to the posterior probability of that Gaussian for the given weight. This is precisely the kind of effect that we are seeking.

Derivatives of the error with respect to the centres of the Gaussians are also easily computed to give

Exercise 5.30

$$\frac{\partial \widetilde{E}}{\partial \mu_j} = \lambda \sum_i \gamma_j(w_i) \frac{(\mu_i - w_j)}{\sigma_j^2} \tag{5.142}$$

which has a simple intuitive interpretation, because it pushes μ_j towards an average of the weight values, weighted by the posterior probabilities that the respective weight parameters were generated by component j. Similarly, the derivatives with respect to the variances are given by

Exercise 5.31

$$\frac{\partial \widetilde{E}}{\partial \sigma_j} = \lambda \sum_i \gamma_j(w_i) \left(\frac{1}{\sigma_j} - \frac{(w_i - \mu_j)^2}{\sigma_j^3} \right) \tag{5.143}$$

which drives σ_j towards the weighted average of the squared deviations of the weights around the corresponding centre μ_j, where the weighting coefficients are again given by the posterior probability that each weight is generated by component j. Note that in a practical implementation, new variables η_j defined by

$$\sigma_j^2 = \exp(\eta_j) \tag{5.144}$$

are introduced, and the minimization is performed with respect to the η_j. This ensures that the parameters σ_j remain positive. It also has the effect of discouraging pathological solutions in which one or more of the σ_j goes to zero, corresponding to a Gaussian component collapsing onto one of the weight parameter values. Such solutions are discussed in more detail in the context of Gaussian mixture models in Section 9.2.1.

For the derivatives with respect to the mixing coefficients π_j, we need to take account of the constraints

$$\sum_j \pi_j = 1, \qquad 0 \leqslant \pi_i \leqslant 1 \tag{5.145}$$

which follow from the interpretation of the π_j as prior probabilities. This can be done by expressing the mixing coefficients in terms of a set of auxiliary variables $\{\eta_j\}$ using the *softmax* function given by

$$\pi_j = \frac{\exp(\eta_j)}{\sum_{k=1}^{M} \exp(\eta_k)}. \tag{5.146}$$

The derivatives of the regularized error function with respect to the $\{\eta_j\}$ then take the form

Exercise 5.32

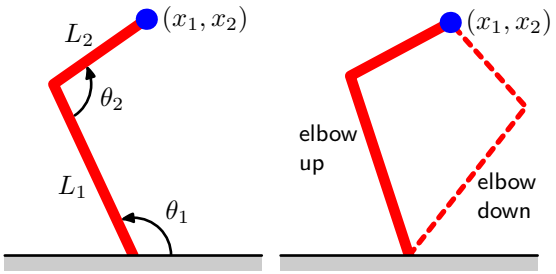


Figure 5.18 The left figure shows a two-link robot arm, in which the Cartesian coordinates (x_1, x_2) of the end effector are determined uniquely by the two joint angles θ_1 and θ_2 and the (fixed) lengths L_1 and L_2 of the arms. This is know as the *forward kinematics* of the arm. In practice, we have to find the joint angles that will give rise to a desired end effector position and, as shown in the right figure, this *inverse kinematics* has two solutions corresponding to 'elbow up' and 'elbow down'.

$$\frac{\partial \widetilde{E}}{\partial \eta_j} = \sum_i \{\pi_j - \gamma_j(w_i)\}\,. \tag{5.147}$$

We see that π_j is therefore driven towards the average posterior probability for component j.

5.6. Mixture Density Networks

The goal of supervised learning is to model a conditional distribution $p(\mathbf{t}|\mathbf{x})$, which for many simple regression problems is chosen to be Gaussian. However, practical machine learning problems can often have significantly non-Gaussian distributions. These can arise, for example, with *inverse problems* in which the distribution can be multimodal, in which case the Gaussian assumption can lead to very poor predictions.

Exercise 5.33

As a simple example of an inverse problem, consider the kinematics of a robot arm, as illustrated in Figure 5.18. The *forward problem* involves finding the end effector position given the joint angles and has a unique solution. However, in practice we wish to move the end effector of the robot to a specific position, and to do this we must set appropriate joint angles. We therefore need to solve the inverse problem, which has two solutions as seen in Figure 5.18.

Forward problems often corresponds to causality in a physical system and generally have a unique solution. For instance, a specific pattern of symptoms in the human body may be caused by the presence of a particular disease. In pattern recognition, however, we typically have to solve an inverse problem, such as trying to predict the presence of a disease given a set of symptoms. If the forward problem involves a many-to-one mapping, then the inverse problem will have multiple solutions. For instance, several different diseases may result in the same symptoms.

In the robotics example, the kinematics is defined by geometrical equations, and the multimodality is readily apparent. However, in many machine learning problems the presence of multimodality, particularly in problems involving spaces of high dimensionality, can be less obvious. For tutorial purposes, however, we shall consider a simple toy problem for which we can easily visualize the multimodality. Data for this problem is generated by sampling a variable x uniformly over the interval $(0, 1)$, to give a set of values $\{x_n\}$, and the corresponding target values t_n are obtained

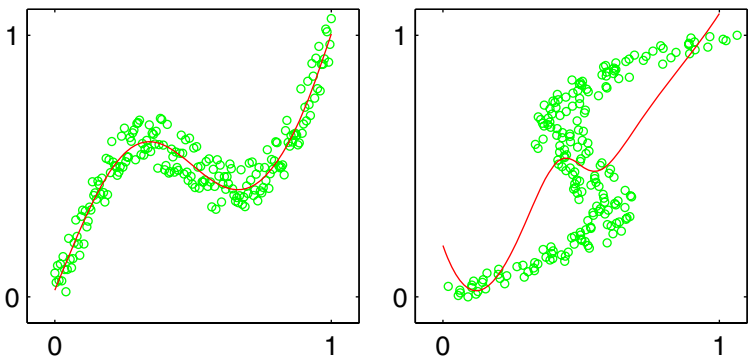


Figure 5.19 On the left is the data set for a simple 'forward problem' in which the red curve shows the result of fitting a two-layer neural network by minimizing the sum-of-squares error function. The corresponding inverse problem, shown on the right, is obtained by exchanging the roles of x and t. Here the same network trained again by minimizing the sum-of-squares error function gives a very poor fit to the data due to the multimodality of the data set.

by computing the function $x_n + 0.3\sin(2\pi x_n)$ and then adding uniform noise over the interval $(-0.1, 0.1)$. The inverse problem is then obtained by keeping the same data points but exchanging the roles of x and t. Figure 5.19 shows the data sets for the forward and inverse problems, along with the results of fitting two-layer neural networks having 6 hidden units and a single linear output unit by minimizing a sum-of-squares error function. Least squares corresponds to maximum likelihood under a Gaussian assumption. We see that this leads to a very poor model for the highly non-Gaussian inverse problem.

We therefore seek a general framework for modelling conditional probability distributions. This can be achieved by using a mixture model for $p(\mathbf{t}|\mathbf{x})$ in which both the mixing coefficients as well as the component densities are flexible functions of the input vector $\mathbf{x}$, giving rise to the *mixture density network*. For any given value of $\mathbf{x}$, the mixture model provides a general formalism for modelling an arbitrary conditional density function $p(\mathbf{t}|\mathbf{x})$. Provided we consider a sufficiently flexible network, we then have a framework for approximating arbitrary conditional distributions.

Here we shall develop the model explicitly for Gaussian components, so that

$$p(\mathbf{t}|\mathbf{x}) = \sum_{k=1}^{K} \pi_k(\mathbf{x}) \mathcal{N}\left(\mathbf{t}|\boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x})\right). \tag{5.148}$$

This is an example of a *heteroscedastic* model since the noise variance on the data is a function of the input vector $\mathbf{x}$. Instead of Gaussians, we can use other distributions for the components, such as Bernoulli distributions if the target variables are binary rather than continuous. We have also specialized to the case of isotropic covariances for the components, although the mixture density network can readily be extended to allow for general covariance matrices by representing the covariances using a Cholesky factorization (Williams, 1996). Even with isotropic components, the conditional distribution $p(\mathbf{t}|\mathbf{x})$ does not assume factorization with respect to the components of $\mathbf{t}$ (in contrast to the standard sum-of-squares regression model) as a consequence of the mixture distribution.

We now take the various parameters of the mixture model, namely the mixing coefficients $\pi_k(\mathbf{x})$, the means $\boldsymbol{\mu}_k(\mathbf{x})$, and the variances $\sigma_k^2(\mathbf{x})$, to be governed by

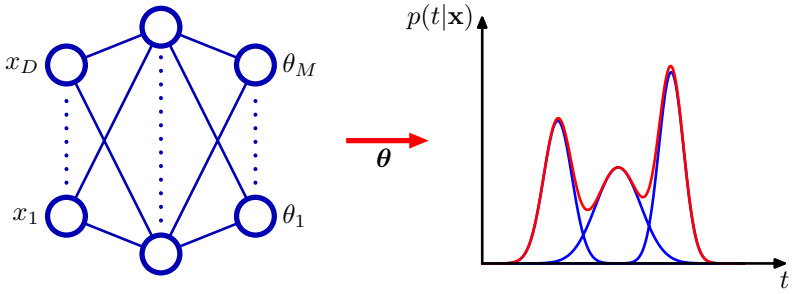


Figure 5.20 The *mixture density network* can represent general conditional probability densities $p(\mathbf{t}|\mathbf{x})$ by considering a parametric mixture model for the distribution of $\mathbf{t}$ whose parameters are determined by the outputs of a neural network that takes $\mathbf{x}$ as its input vector.

the outputs of a conventional neural network that takes $\mathbf{x}$ as its input. The structure of this mixture density network is illustrated in Figure 5.20. The mixture density network is closely related to the mixture of experts discussed in Section 14.5.3. The principle difference is that in the mixture density network the same function is used to predict the parameters of all of the component densities as well as the mixing coefficients, and so the nonlinear hidden units are shared amongst the input-dependent functions.

The neural network in Figure 5.20 can, for example, be a two-layer network having sigmoidal ('tanh') hidden units. If there are L components in the mixture model (5.148), and if $\mathbf{t}$ has K components, then the network will have L output unit activations denoted by a_k^π that determine the mixing coefficients $\pi_k(\mathbf{x})$, K outputs denoted by a_k^σ that determine the kernel widths $\sigma_k(\mathbf{x})$, and $L \times K$ outputs denoted by a_{kj}^μ that determine the components $\mu_{kj}(\mathbf{x})$ of the kernel centres $\boldsymbol{\mu}_k(\mathbf{x})$. The total number of network outputs is given by $(K+2)L$, as compared with the usual K outputs for a network, which simply predicts the conditional means of the target variables.

The mixing coefficients must satisfy the constraints

$$\sum_{k=1}^{K} \pi_k(\mathbf{x}) = 1, \qquad 0 \leqslant \pi_k(\mathbf{x}) \leqslant 1 \tag{5.149}$$

which can be achieved using a set of softmax outputs

$$\pi_k(\mathbf{x}) = \frac{\exp(a_k^\pi)}{\sum_{l=1}^{K} \exp(a_l^\pi)}. \tag{5.150}$$

Similarly, the variances must satisfy $\sigma_k^2(\mathbf{x}) \geqslant 0$ and so can be represented in terms of the exponentials of the corresponding network activations using

$$\sigma_k(\mathbf{x}) = \exp(a_k^\sigma). \tag{5.151}$$

Finally, because the means $\boldsymbol{\mu}_k(\mathbf{x})$ have real components, they can be represented

directly by the network output activations

$$\mu_{kj}(\mathbf{x}) = a_{kj}^{\mu}. \tag{5.152}$$

The adaptive parameters of the mixture density network comprise the vector $\mathbf{w}$ of weights and biases in the neural network, that can be set by maximum likelihood, or equivalently by minimizing an error function defined to be the negative logarithm of the likelihood. For independent data, this error function takes the form

$$E(\mathbf{w}) = -\sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{k} \pi_k(\mathbf{x}_n, \mathbf{w}) \mathcal{N}\left(\mathbf{t}_n | \boldsymbol{\mu}_k(\mathbf{x}_n, \mathbf{w}), \sigma_k^2(\mathbf{x}_n, \mathbf{w})\right) \right\} \tag{5.153}$$

where we have made the dependencies on $\mathbf{w}$ explicit.

In order to minimize the error function, we need to calculate the derivatives of the error $E(\mathbf{w})$ with respect to the components of $\mathbf{w}$. These can be evaluated by using the standard backpropagation procedure, provided we obtain suitable expressions for the derivatives of the error with respect to the output-unit activations. These represent error signals δ for each pattern and for each output unit, and can be backpropagated to the hidden units and the error function derivatives evaluated in the usual way. Because the error function (5.153) is composed of a sum of terms, one for each training data point, we can consider the derivatives for a particular pattern n and then find the derivatives of E by summing over all patterns.

Because we are dealing with mixture distributions, it is convenient to view the mixing coefficients $\pi_k(\mathbf{x})$ as $\mathbf{x}$-dependent prior probabilities and to introduce the corresponding posterior probabilities given by

$$\gamma_k(\mathbf{t}|\mathbf{x}) = \frac{\pi_k \mathcal{N}_{nk}}{\sum_{l=1}^{K} \pi_l \mathcal{N}_{nl}} \tag{5.154}$$

where $\mathcal{N}_{nk}$ denotes $\mathcal{N}\left(\mathbf{t}_n | \boldsymbol{\mu}_k(\mathbf{x}_n), \sigma_k^2(\mathbf{x}_n)\right)$.

Exercise 5.34 The derivatives with respect to the network output activations governing the mixing coefficients are given by

$$\frac{\partial E_n}{\partial a_k^{\pi}} = \pi_k - \gamma_k. \tag{5.155}$$

Exercise 5.35 Similarly, the derivatives with respect to the output activations controlling the component means are given by

$$\frac{\partial E_n}{\partial a_{kl}^{\mu}} = \gamma_k \left\{ \frac{\mu_{kl} - t_l}{\sigma_k^2} \right\}. \tag{5.156}$$

Exercise 5.36 Finally, the derivatives with respect to the output activations controlling the component variances are given by

$$\frac{\partial E_n}{\partial a_k^{\sigma}} = -\gamma_k \left\{ \frac{\|\mathbf{t} - \boldsymbol{\mu}_k\|^2}{\sigma_k^3} - \frac{1}{\sigma_k} \right\}. \tag{5.157}$$

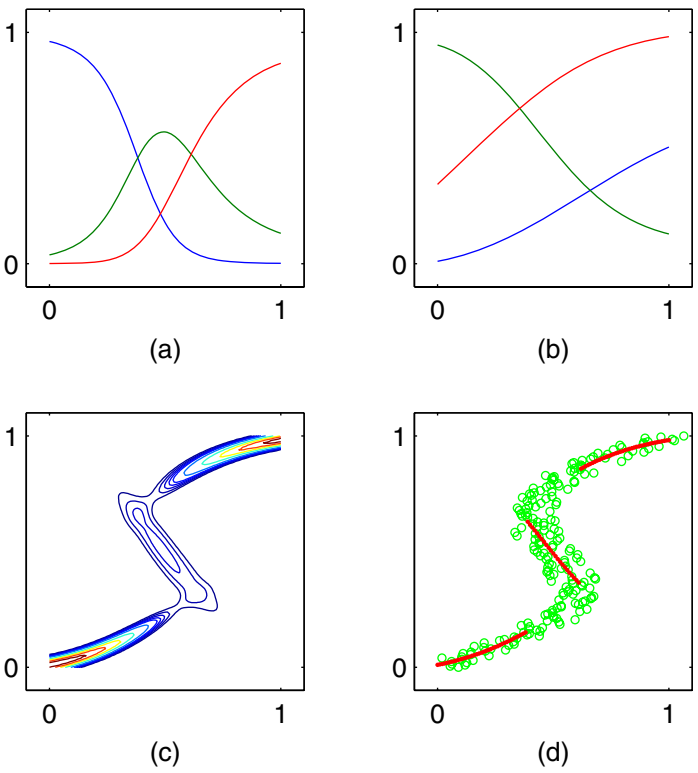


Figure 5.21 (a) Plot of the mixing coefficients $\pi_k(x)$ as a function of x for the three kernel functions in a mixture density network trained on the data shown in Figure 5.19. The model has three Gaussian components, and uses a two-layer multilayer perceptron with five 'tanh' sigmoidal units in the hidden layer, and nine outputs (corresponding to the 3 means and 3 variances of the Gaussian components and the 3 mixing coefficients). At both small and large values of x, where the conditional probability density of the target data is unimodal, only one of the kernels has a high value for its prior probability, while at intermediate values of x, where the conditional density is trimodal, the three mixing coefficients have comparable values. (b) Plots of the means $\mu_k(x)$ using the same colour coding as for the mixing coefficients. (c) Plot of the contours of the corresponding conditional probability density of the target data for the same mixture density network. (d) Plot of the approximate conditional mode, shown by the red points, of the conditional density.

We illustrate the use of a mixture density network by returning to the toy example of an inverse problem shown in Figure 5.19. Plots of the mixing coefficients $\pi_k(x)$, the means $\mu_k(x)$, and the conditional density contours corresponding to $p(t|x)$, are shown in Figure 5.21. The outputs of the neural network, and hence the parameters in the mixture model, are necessarily continuous single-valued functions of the input variables. However, we see from Figure 5.21(c) that the model is able to produce a conditional density that is unimodal for some values of x and trimodal for other values by modulating the amplitudes of the mixing components $\pi_k(\mathbf{x})$.

Once a mixture density network has been trained, it can predict the conditional density function of the target data for any given value of the input vector. This conditional density represents a complete description of the generator of the data, so far as the problem of predicting the value of the output vector is concerned. From this density function we can calculate more specific quantities that may be of interest in different applications. One of the simplest of these is the mean, corresponding to the conditional average of the target data, and is given by

$$\mathbb{E}\left[\mathbf{t}|\mathbf{x}\right] = \int \mathbf{t} p(\mathbf{t}|\mathbf{x})\, \mathrm{d}\mathbf{t} = \sum_{k=1}^{K} \pi_k(\mathbf{x}) \boldsymbol{\mu}_k(\mathbf{x}) \tag{5.158}$$

where we have used (5.148). Because a standard network trained by least squares is approximating the conditional mean, we see that a mixture density network can reproduce the conventional least-squares result as a special case. Of course, as we have already noted, for a multimodal distribution the conditional mean is of limited value.

We can similarly evaluate the variance of the density function about the conditional average, to give

Exercise 5.37

$$s^2(\mathbf{x}) = \mathbb{E}\left[\|\mathbf{t} - \mathbb{E}[\mathbf{t}|\mathbf{x}]\|^2 \,|\mathbf{x}\right] \tag{5.159}$$

$$= \sum_{k=1}^{K} \pi_k(\mathbf{x}) \left\{ \sigma_k^2(\mathbf{x}) + \left\| \boldsymbol{\mu}_k(\mathbf{x}) - \sum_{l=1}^{K} \pi_l(\mathbf{x}) \boldsymbol{\mu}_l(\mathbf{x}) \right\|^2 \right\} \tag{5.160}$$

where we have used (5.148) and (5.158). This is more general than the corresponding least-squares result because the variance is a function of $\mathbf{x}$.

We have seen that for multimodal distributions, the conditional mean can give a poor representation of the data. For instance, in controlling the simple robot arm shown in Figure 5.18, we need to pick one of the two possible joint angle settings in order to achieve the desired end-effector location, whereas the average of the two solutions is not itself a solution. In such cases, the conditional mode may be of more value. Because the conditional mode for the mixture density network does not have a simple analytical solution, this would require numerical iteration. A simple alternative is to take the mean of the most probable component (i.e., the one with the largest mixing coefficient) at each value of $\mathbf{x}$. This is shown for the toy data set in Figure 5.21(d).

5.7. Bayesian Neural Networks

So far, our discussion of neural networks has focussed on the use of maximum likelihood to determine the network parameters (weights and biases). Regularized maximum likelihood can be interpreted as a MAP (maximum posterior) approach in which the regularizer can be viewed as the logarithm of a prior parameter distribution. However, in a Bayesian treatment we need to marginalize over the distribution of parameters in order to make predictions.

In Section 3.3, we developed a Bayesian solution for a simple linear regression model under the assumption of Gaussian noise. We saw that the posterior distribution, which is Gaussian, could be evaluated exactly and that the predictive distribution could also be found in closed form. In the case of a multilayered network, the highly nonlinear dependence of the network function on the parameter values means that an exact Bayesian treatment can no longer be found. In fact, the log of the posterior distribution will be nonconvex, corresponding to the multiple local minima in the error function.

The technique of variational inference, to be discussed in Chapter 10, has been applied to Bayesian neural networks using a factorized Gaussian approximation

to the posterior distribution (Hinton and van Camp, 1993) and also using a full-covariance Gaussian (Barber and Bishop, 1998a; Barber and Bishop, 1998b). The most complete treatment, however, has been based on the Laplace approximation (MacKay, 1992c; MacKay, 1992b) and forms the basis for the discussion given here. We will approximate the posterior distribution by a Gaussian, centred at a mode of the true posterior. Furthermore, we shall assume that the covariance of this Gaussian is small so that the network function is approximately linear with respect to the parameters over the region of parameter space for which the posterior probability is significantly nonzero. With these two approximations, we will obtain models that are analogous to the linear regression and classification models discussed in earlier chapters and so we can exploit the results obtained there. We can then make use of the evidence framework to provide point estimates for the hyperparameters and to compare alternative models (for example, networks having different numbers of hidden units). To start with, we shall discuss the regression case and then later consider the modifications needed for solving classification tasks.

5.7.1 Posterior parameter distribution

Consider the problem of predicting a single continuous target variable t from a vector $\mathbf{x}$ of inputs (the extension to multiple targets is straightforward). We shall suppose that the conditional distribution $p(t|\mathbf{x})$ is Gaussian, with an $\mathbf{x}$-dependent mean given by the output of a neural network model $y(\mathbf{x}, \mathbf{w})$, and with precision (inverse variance) β

$$p(t|\mathbf{x}, \mathbf{w}, \beta) = \mathcal{N}(t|y(\mathbf{x}, \mathbf{w}), \beta^{-1}). \tag{5.161}$$

Similarly, we shall choose a prior distribution over the weights $\mathbf{w}$ that is Gaussian of the form

$$p(\mathbf{w}|\alpha) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}). \tag{5.162}$$

For an i.i.d. data set of N observations $\mathbf{x}_1, \ldots, \mathbf{x}_N$, with a corresponding set of target values $\mathcal{D} = \{t_1, \ldots, t_N\}$, the likelihood function is given by

$$p(\mathcal{D}|\mathbf{w}, \beta) = \prod_{n=1}^{N} \mathcal{N}(t_n|y(\mathbf{x}_n, \mathbf{w}), \beta^{-1}) \tag{5.163}$$

and so the resulting posterior distribution is then

$$p(\mathbf{w}|\mathcal{D}, \alpha, \beta) \propto p(\mathbf{w}|\alpha)p(\mathcal{D}|\mathbf{w}, \beta). \tag{5.164}$$

which, as a consequence of the nonlinear dependence of $y(\mathbf{x}, \mathbf{w})$ on $\mathbf{w}$, will be non-Gaussian.

We can find a Gaussian approximation to the posterior distribution by using the Laplace approximation. To do this, we must first find a (local) maximum of the posterior, and this must be done using iterative numerical optimization. As usual, it is convenient to maximize the logarithm of the posterior, which can be written in the

form

$$\ln p(\mathbf{w}|\mathcal{D}) = -\frac{\alpha}{2}\mathbf{w}^{\mathrm{T}}\mathbf{w} - \frac{\beta}{2}\sum_{n=1}^{N}\{y(\mathbf{x}_n, \mathbf{w}) - t_n\}^2 + \text{const} \tag{5.165}$$

which corresponds to a regularized sum-of-squares error function. Assuming for the moment that α and β are fixed, we can find a maximum of the posterior, which we denote $\mathbf{w}_{\mathrm{MAP}}$, by standard nonlinear optimization algorithms such as conjugate gradients, using error backpropagation to evaluate the required derivatives.

Having found a mode $\mathbf{w}_{\mathrm{MAP}}$, we can then build a local Gaussian approximation by evaluating the matrix of second derivatives of the negative log posterior distribution. From (5.165), this is given by

$$\mathbf{A} = -\nabla\nabla \ln p(\mathbf{w}|\mathcal{D}, \alpha, \beta) = \alpha\mathbf{I} + \beta\mathbf{H} \tag{5.166}$$

where $\mathbf{H}$ is the Hessian matrix comprising the second derivatives of the sum-of-squares error function with respect to the components of $\mathbf{w}$. Algorithms for computing and approximating the Hessian were discussed in Section 5.4. The corresponding Gaussian approximation to the posterior is then given from (4.134) by

$$q(\mathbf{w}|\mathcal{D}) = \mathcal{N}(\mathbf{w}|\mathbf{w}_{\mathrm{MAP}}, \mathbf{A}^{-1}). \tag{5.167}$$

Similarly, the predictive distribution is obtained by marginalizing with respect to this posterior distribution

$$p(t|\mathbf{x}, \mathcal{D}) = \int p(t|\mathbf{x}, \mathbf{w})q(\mathbf{w}|\mathcal{D})\,\mathrm{d}\mathbf{w}. \tag{5.168}$$

However, even with the Gaussian approximation to the posterior, this integration is still analytically intractable due to the nonlinearity of the network function $y(\mathbf{x}, \mathbf{w})$ as a function of $\mathbf{w}$. To make progress, we now assume that the posterior distribution has small variance compared with the characteristic scales of $\mathbf{w}$ over which $y(\mathbf{x}, \mathbf{w})$ is varying. This allows us to make a Taylor series expansion of the network function around $\mathbf{w}_{\mathrm{MAP}}$ and retain only the linear terms

$$y(\mathbf{x}, \mathbf{w}) \simeq y(\mathbf{x}, \mathbf{w}_{\mathrm{MAP}}) + \mathbf{g}^{\mathbf{T}}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}) \tag{5.169}$$

where we have defined

$$\mathbf{g} = \nabla_{\mathbf{w}} y(\mathbf{x}, \mathbf{w})|_{\mathbf{w}=\mathbf{w}_{\mathrm{MAP}}}. \tag{5.170}$$

With this approximation, we now have a linear-Gaussian model with a Gaussian distribution for $p(\mathbf{w})$ and a Gaussian for $p(t|\mathbf{w})$ whose mean is a linear function of $\mathbf{w}$ of the form

$$p(t|\mathbf{x}, \mathbf{w}, \beta) \simeq \mathcal{N}\left(t|y(\mathbf{x}, \mathbf{w}_{\mathrm{MAP}}) + \mathbf{g}^{\mathbf{T}}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}}), \beta^{-1}\right). \tag{5.171}$$

Exercise 5.38

We can therefore make use of the general result (2.115) for the marginal $p(t)$ to give

$$p(t|\mathbf{x}, \mathcal{D}, \alpha, \beta) = \mathcal{N}\left(t|y(\mathbf{x}, \mathbf{w}_{\mathrm{MAP}}), \sigma^2(\mathbf{x})\right) \tag{5.172}$$